Politecnico di Milano
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Bayesian Learning and Monte Carlo Simulation

Final project: Yacht Hydrodynamic Resistance

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1 Data and Problem

We analyse the *Yacht Hydrodynamics* dataset from the UCI Machine Learning Repository [1, 2]: 22 sailing-yacht hull forms tested at several speeds, for $n = 308$ observations. Each row holds five hull *form factors* — `LCB`, `prismatic`, `L_disp`, `beam_draught` and `L_beam` (constant within a hull) — and the dimensionless **Froude number** $Fn = v/\sqrt{gL} \in [0.125, 0.45]$; the response is the residuary resistance per unit weight of displacement (Table 2).

For slender hulls, physics predicts a power law resistance $\propto Fn^{\gamma_F}$ with $\gamma_F \approx 4$. We study it in a Bayesian framework:

1. log–log regression to estimate and *verify* $\hat{\beta}_F \approx 4$, with a prior-sensitivity check;
2. BAS/BMA over the 2^5 hull-factor submodels, to see which carry signal once Fn is included;
3. a heteroscedasticity-corrected nonlinear model on the retained factors, contrasted with the log–log fit;
4. posterior predictive curves, with and without the heteroscedasticity treatment, against the data.

2 Log–log Regression

2.1 Model and prior

Taking logarithms turns the physical power law resistance $\propto Fn^{\gamma_F}$ into a linear model in which the exponent appears directly as a regression slope. We therefore fit

$$\log(\text{resistance}) = \beta_0 + \beta_F \, \log(\text{Froude}) + \sum_{k=1}^{5} \beta_k \, \text{hull}_k + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad (1)$$

where hull_k denotes the five geometric form factors; the smallest observed resistance is $0.01 > 0$, so the logarithm is well defined for every record and no offset is required. As a baseline we adopt the non-informative **Jeffreys reference prior**

$$\pi(\boldsymbol{\beta} \mid \sigma^2) \;\propto\; 1, \qquad \pi(\sigma^2) \;\propto\; \sigma^{-2}, \quad (2)$$

which lets the data dominate the inference; the sensitivity of the conclusions to an informative prior is examined in section 2.4.

2.2 Posterior and credible intervals

With the Jeffreys reference prior in Eq. (2), the Gaussian linear model has a closed-form posterior. Let $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$. Then

$$\boldsymbol{\beta} \mid \sigma^2, \boldsymbol{y} \sim \mathcal{N}\Big(\hat{\boldsymbol{\beta}}, \, \sigma^2 \, (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\Big), \quad (3)$$

Integrating out σ^2 gives a multivariate Student-t posterior for $\boldsymbol{\beta}$ with $n - p$ degrees of freedom, so the 95% credible intervals coincide numerically with the classical OLS confidence intervals.

2.3 Results

Table 3 (in the appendix) reports the posterior summaries. The model explains the data very well ($R^2 = 0.97$, residual standard deviation $s = 0.32$ on the log scale) and the posterior of the Froude exponent is sharply concentrated,

$$\beta_F \approx 4.69, \qquad 95\%\ \text{CI} = [4.60,\ 4.78],$$

confirming the expected power-law behaviour resistance $\sim Fn^4$ and verifying the project requirement $\hat{\beta}_F \approx 4$.

Fig. 2 (appendix) shows the log–log scatter with the posterior-mean line and its very tight 95% credible band (left), and the posterior of β_F (right): the whole posterior mass sits above the nominal value 4 ($\Pr(\beta_F > 4 \mid \boldsymbol{y}) \approx 1$), so the empirical exponent is best described as *slightly steeper* than the textbook value of four.

2.4 Sensitivity check: independent informative priors

The reference-prior analysis *verifies* $\beta_F \approx 4$ a posteriori; a complementary check *encodes* the physics in the prior and asks how strongly the prior must insist on 4 before it overrides the data. We replace Eq. (2) with **independent proper priors**,

$$\beta_F \sim \mathcal{N}(4, \tau^2), \qquad \beta_j \sim \mathcal{N}(0, 100^2)\ \ (j \neq F), \qquad \sigma^2 \sim \text{Inv-Gamma}(10^{-3}, 10^{-3}), \tag{4}$$

all components mutually independent. Since $\boldsymbol{\beta}$ and σ^2 are now independent a priori, unlike in the conjugate Normal–Inverse-Gamma family, the joint posterior is no longer available in closed form, but both full conditionals remain standard,

$$\boldsymbol{\beta} \mid \sigma^2, \boldsymbol{y} \sim \mathcal{N}\big(\boldsymbol{V}(\boldsymbol{S}_0^{-1}\boldsymbol{m}_0 + \boldsymbol{X}^\top \boldsymbol{y}/\sigma^2),\ \boldsymbol{V}\big), \qquad \boldsymbol{V} = (\boldsymbol{S}_0^{-1} + \boldsymbol{X}^\top \boldsymbol{X}/\sigma^2)^{-1}, \tag{5}$$

$$\sigma^2 \mid \boldsymbol{\beta}, \boldsymbol{y} \sim \text{Inv-Gamma}\big(a_0 + \tfrac{n}{2},\ b_0 + \tfrac{1}{2}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2\big), \tag{6}$$

where $\boldsymbol{m}_0$ and $\boldsymbol{S}_0$ collect the prior means and variances of Eq. (4). We therefore sample the posterior with a **Gibbs sampler** (10 000 iterations, 2 000 burn-in; the effective sample size for β_F exceeds 4 000 in every run) and sweep the prior strength $\tau \in \{0.05, 0.1, 0.2, 0.5\}$.

Table 4 (appendix) summarises the sweep; the posterior densities are overlaid in Fig. 5 (appendix). A moderately informative prior ($\tau = 0.5$) reproduces the reference result. Tightening the prior pulls the posterior mean towards 4, but only the near-dogmatic $\tau = 0.05$ (prior 95% interval $\approx [3.9, 4.1]$) drags it as far as 4.33, and even then $\Pr(\beta_F > 4 \mid \boldsymbol{y}) = 1$.

For any moderately diffuse physical prior ($\tau \geq 0.1$) the likelihood dominates and the posterior mean stays in 4.56–4.68. The conclusion that the empirical exponent is genuinely steeper than Fn^4 is therefore **robust to the prior**: it is a feature of the $n = 308$ observations, not an artefact of the non-informative choice, a transparent instance of prior–data conflict, in which an informative likelihood overwhelms all but an unreasonably confident prior.

3 BAS/BMA model selection over hull covariates

The first part establishes the dominant physical relationship: after taking logs, the resistance is essentially a power law of the Froude number. The second question is conditional on that fact: once $\log(Fn)$ is already in the model, which hull-geometry variables still carry independent signal?

We address this with Bayesian model averaging using `bas.lm`. The model space contains the five optional geometry covariates `LCB`, `prismatic`, `L_disp`, `beam_draught` and `L_beam`. Therefore the number of candidate submodels is only $2^5 = 32$. The term $\log(Fn)$ is forced into every model, because the project is not asking whether speed matters; it is asking which hull covariates matter *after* speed has been accounted for. In the R code this is enforced with `include.always = ~ log_Fr`, while the five hull terms are left optional.

The main diagnostic is the posterior inclusion probability (PIP). A PIP is the posterior probability that a covariate is included in the true regression model, after averaging over all 32 submodels. Hence a large PIP indicates that the variable repeatedly appears in high-posterior-probability models, while a small PIP means that the data do not strongly need that variable once the other terms are considered.

Using the BIC approximation to the marginal likelihood, the highest posterior model is the one containing $\log(Fn)$ and `beam_draught`, but its posterior probability is not dominant. The approximate PIPs are reported in Table 1. The largest value is for `beam_draught`, but even there the evidence is only moderate. The remaining covariates have weak inclusion probabilities. Thus the BAS/BMA conclusion is deliberately cautious: the Froude power law is the stable signal; the hull-form variables may have secondary effects, but no individual hull covariate is strongly identified by the homoscedastic log–log regression.

This agrees with the full-model coefficient intervals in Table 3, where all five hull coefficients have 95% credible intervals crossing zero. BAS/BMA strengthens that interpretation because it checks the whole model space instead of relying on one full regression model.

Table 1: Approximate posterior inclusion probabilities for the five optional hull-geometry covariates when $\log(Fn)$ is forced into every model. Values are based on enumeration of the $2^5 = 32$ submodels using a BIC marginal-likelihood approximation, matching the `prior = "BIC"` specification in `bas.lm`.

Covariate	**PIP**	**Interpretation**
`beam_draught`	0.57	moderate, not decisive
`prismatic`	0.27	weak
`LCB`	0.21	weak
`L_disp`	0.21	weak
`L_beam`	0.19	weak

4 Heteroscedasticity-Corrected Nonlinear Model

5 Posterior Predictive Curves

The final comparison is made on the original resistance scale, because this is where the data are observed and where the two error assumptions imply visibly different uncertainty bands. For a fair comparison, both curves in Fig. 1 are posterior predictive curves: each posterior draw includes uncertainty in the mean curve and one new observation-level error term. The observed yacht-resistance data are overlaid on the same scale.

For the log–log regression, the posterior predictive distribution is lognormal after transforming back to resistance. After drawing $(\boldsymbol{\beta}, \sigma^2)$ from the posterior, replicated values are generated as

$$\log y_i^{\text{rep}} \sim \mathcal{N}(x_i^\top \boldsymbol{\beta}, \sigma^2), \qquad y_i^{\text{rep}} = \exp(\log y_i^{\text{rep}}).$$

For the heteroscedastic nonlinear M1 model, replicated values are generated on the original scale with a mean-dependent standard deviation,

$$y_i^{\text{rep}} \sim \mathcal{N}(\mu_i, \sigma_i^2), \qquad \mu_i = \exp\{b_0 + \gamma_F \widetilde{\log Fn}_i + \beta_B \widetilde{bd}_i\}, \qquad \sigma_i = \sigma_0 \mu_i^\delta.$$

The displayed curves hold the hull form fixed at a representative value: `beam_draught` is set to its sample median, while the other log–log hull covariates are set to their sample means. This isolates the Froude–resistance relationship instead of mixing different hull geometries.

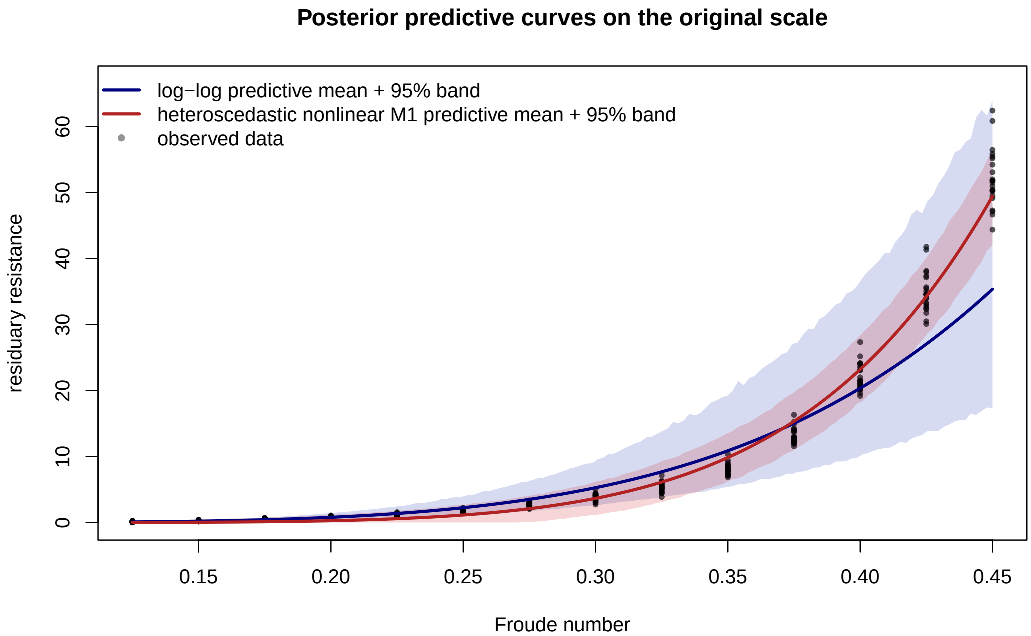


Figure 1: Posterior predictive curves on the original resistance scale. The blue band is the log–log model after back-transformation, and the red band is the heteroscedastic nonlinear M1 model. Points are the observed yacht resistance data.

The log–log fit keeps the Froude exponent close to the physical fourth-power law and produces multiplicative uncertainty. Its band is tight near zero resistance and expands as resistance grows, which describes the low and middle Froude range reasonably well. However, the mean curve is too flat at the highest Froude values, where the observed resistance rises faster than the fitted $Fn^{4.69}$ relationship.

The heteroscedastic nonlinear fit behaves differently. Its mean curve is steeper ($\gamma_F \approx 6.4$) and follows the high-Froude observations more closely. Because its variance is learned through $\sigma_i = \sigma_0 \mu_i^\delta$ with $\delta \approx 0.4$, the predictive band is not forced to have constant width. It widens where resistance and scatter are larger and tightens toward the low-Froude region. This is the practical payoff of the heteroscedasticity treatment: the uncertainty band adapts to the data rather than imposing a single residual scale across the whole Froude range.

Overall, the posterior predictive curves show that the two models differ mainly in how they scale uncertainty with resistance. The log–log model gives a useful positive predictive distribution, but it imposes multiplicative uncertainty and misses part of the high-speed curvature. The heteroscedastic nonlinear model learns the variance growth directly, tracks the observed data more closely, and supports the conclusion that the treatment of heteroscedasticity is essential for interpreting both the Froude exponent and the selected hull covariate `beam_draught`.

References

[1] I. Ortigosa, R. Lopez, and J. Garcia. *Yacht Hydrodynamics*. UCI Machine Learning Repository. DOI: 10.24432/C5XG7R. 2007.

[2] J. Gerritsma, R. Onnink, and A. Versluis. *Geometry, Resistance and Stability of the Delft Systematic Yacht Hull Series*. Technical Report. Delft University of Technology, Ship Hydromechanics Laboratory, 1981.

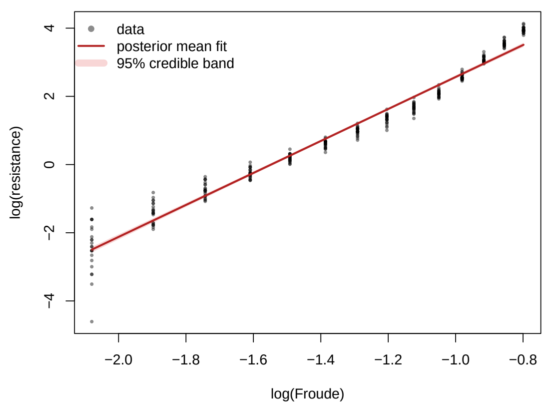
A Tables and Figures

Table 2: Variables of the yacht hydrodynamics dataset.

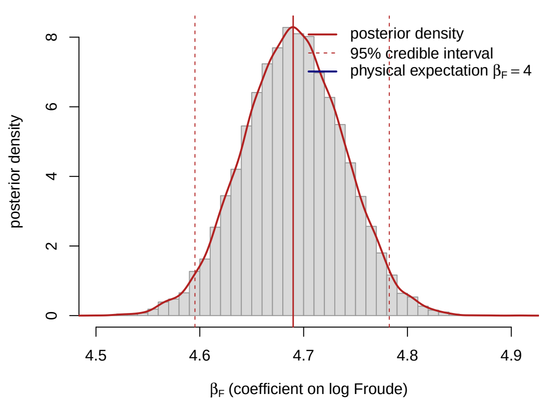
Variable	Description
`LCB`	Longitudinal position of the centre of buoyancy
`prismatic`	Prismatic coefficient
`L_disp`	Length–displacement ratio
`beam_draught`	Beam–draught ratio
`L_beam`	Length–beam ratio
`Froude`	Froude number, $Fn = v/\sqrt{gL}$
`resistance`	Residuary resistance per unit weight of displacement

Table 3: Posterior summary of the log–log model in Eq. (1) under the reference prior. Point estimates and 95% credible intervals coincide with the OLS fit (`lm`/`confint`); Pr(> 0) is the posterior probability that the coefficient is positive.

Coefficient	Mean	SD	2.5%	97.5%	Pr(> 0)
Intercept β_0	7.183	0.983	5.248	9.119	1.00
log(Froude) β_F	4.690	0.048	4.596	4.785	1.00
`LCB`	0.022	0.012	−0.003	0.046	0.96
`prismatic`	−0.621	1.601	−3.771	2.529	0.35
`L_disp`	0.465	0.513	−0.546	1.475	0.82
`beam_draught`	−0.066	0.200	−0.460	0.327	0.37
`L_beam`	−0.464	0.515	−1.477	0.549	0.18



(a) Power-law fit on log–log axes.



(b) Posterior of β_F.

Figure 2: Left: observed log(resistance) against log(Froude) with the posterior-mean regression line (hull covariates held at their means) and its 95% credible band. Right: Student-t posterior density of the Froude exponent β_F; the dashed lines mark the 95% credible interval and the solid blue line the physical reference value $\beta_F = 4$.

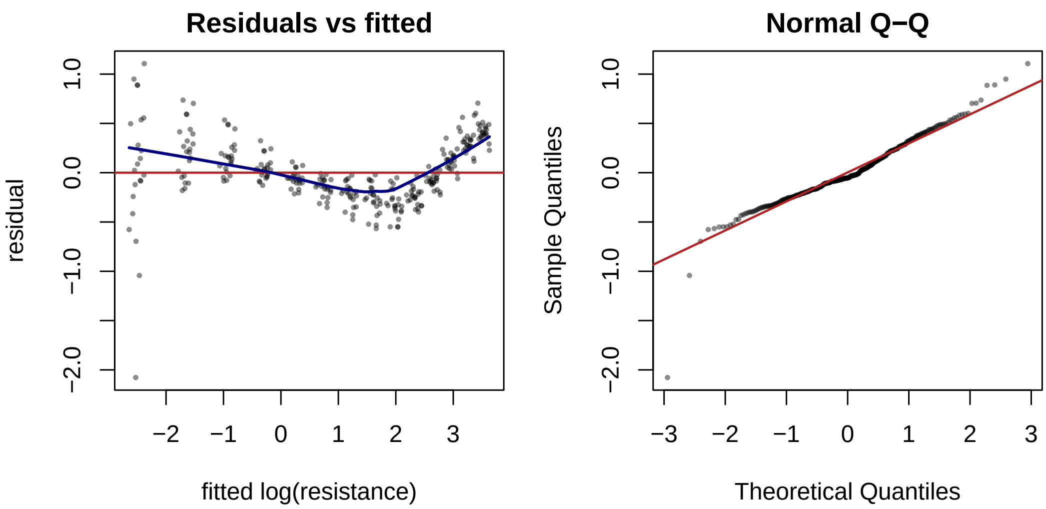


Figure 3: Residual diagnostics for the log–log model. Left: residuals versus fitted values, with a LOWESS trend (blue) revealing speed-dependent curvature and a variance that shrinks as the fitted value grows. Right: Normal Q–Q plot, showing heavier-than-Gaussian lower tails.

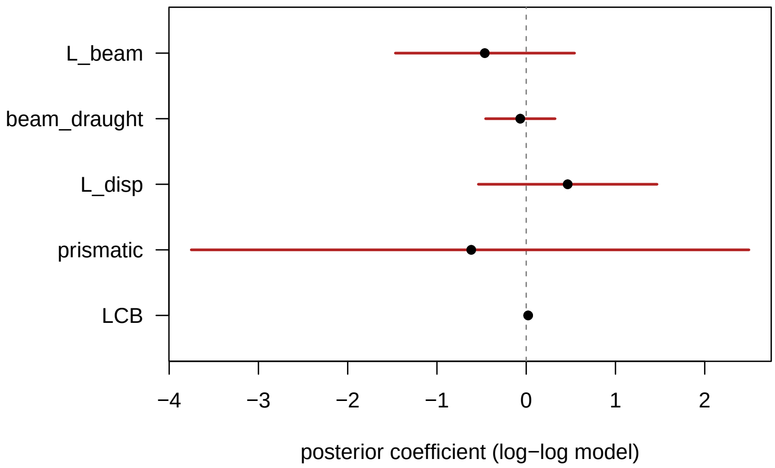


Figure 4: Posterior means and 95% credible intervals of the five hull-geometry coefficients in the log–log model. Every interval straddles zero (dashed line), indicating that no single form factor is resolved once the Froude number is accounted for.

Hull-geometry covariates. Once the Froude number is in the model, none of the five hull form factors is individually credible: every 95% interval comfortably contains zero (Table 3 and Fig. 4). The form factors vary only across the 22 hulls and are strongly collinear — the `prismatic` coefficient, for instance, has a posterior standard deviation of about 1.6. This near-redundancy is precisely what the BAS/BMA model selection in section 3 resolves, through posterior inclusion probabilities rather than single-model p-values.

Residual diagnostics. Although the global fit is excellent, the residuals (Fig. 3) reveal two systematic defects. First, a ∪-shaped trend against the fitted values: a single constant exponent slightly over-predicts at intermediate speeds and under-predicts at the highest speeds. Second, the residual spread is markedly larger at low fitted values, i.e. at low Froude numbers where the resistance is tiny and dominated by measurement noise — a textbook signature of heteroscedasticity, also visible as the fanning of the data cloud on the left of Fig. 2. Both findings directly motivate the heteroscedasticity-corrected nonlinear model developed in the main text.

Table 4: Posterior of the Froude exponent β_F under the independent informative priors in Eq. (4) of increasing strength, obtained by Gibbs sampling, compared with the closed-form reference-prior result.

Prior on β_F	**Mean**	**2.5%**	**97.5%**	$\Pr(\beta_F > 4)$
$\mathcal{N}(4, 0.05^2)$	4.33	4.25	4.41	1.00
$\mathcal{N}(4, 0.10^2)$	4.56	4.47	4.64	1.00
$\mathcal{N}(4, 0.20^2)$	4.65	4.56	4.75	1.00
$\mathcal{N}(4, 0.50^2)$	4.68	4.59	4.78	1.00
reference $\pi(\boldsymbol{\beta}, \sigma^2) \propto \sigma^{-2}$	4.69	4.60	4.78	1.00

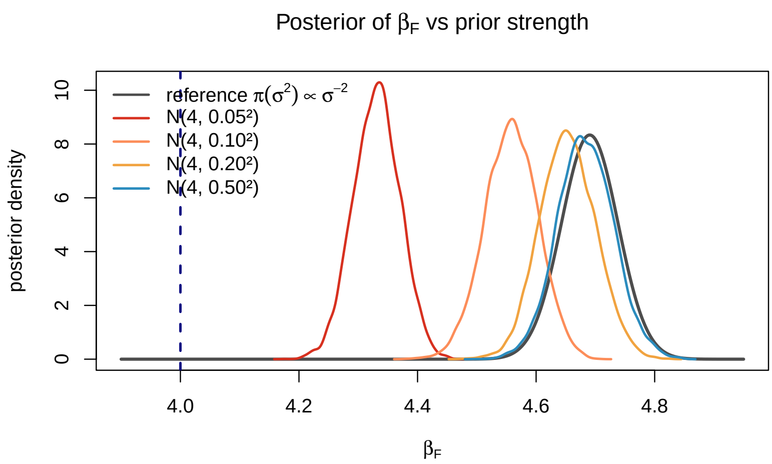


Figure 5: Posterior density of the Froude exponent β_F under the independent informative priors $\mathcal{N}(4, \tau^2)$ of increasing strength (coloured, Gibbs samples) and under the reference prior (grey, closed-form Student-t). The dashed blue line marks the physical value $\beta_F = 4$. Tightening the prior toward 4 shifts the posterior only modestly; the data keep it well above 4 unless the prior is unreasonably confident.